

TM Forum Component

Work Order Management

TMFC061

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Notice

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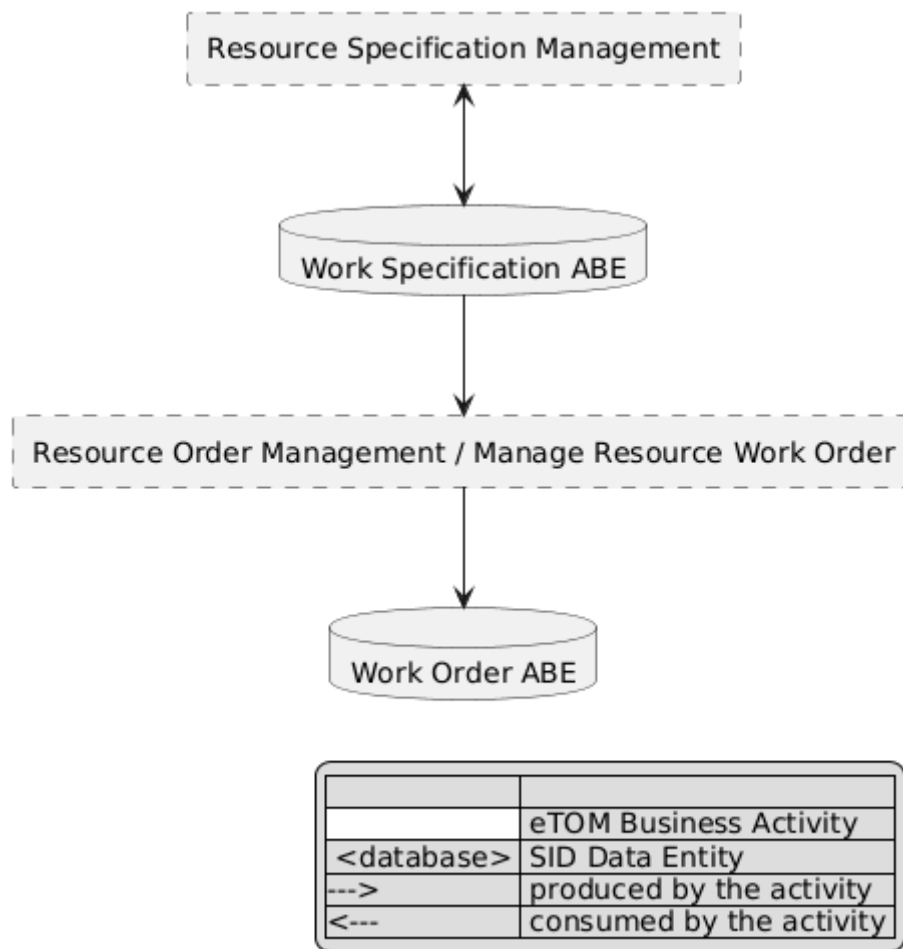
TMFC061 – Work Order Management

1. Overview

Component Name	ID	Description	ODA Function Block
Work Order Management	TMFC061	Work Order Management component is responsible of the setup and maintenance requests from initiation to completion. Work Order is a formal request that outlines essential details related to necessary setup/maintenance tasks. These details typically include Nature of the task A description of the work that needs to be done. Location Where the maintenance or repair is required. Required materials The items or resources needed for the job. Assigned personnel The technician or contractor responsible for executing the work. Timeline for completion The expected timeframe within which the task should be finished.	Production

2. eTOM Processes, SID Data Entities and Functional Framework Functions

2.1. eTOM L2 - SID ABEs links



2.2. eTOM business activities

eTOM business activities this ODA Component is responsible for:

Identifier	Level	Business Activity Name	Description
1.5.5	L2	Resource Order Management	Resource Order Management business process directs and controls ordering, scheduling, and allocation of resources (such as materials, equipment, and personnel) within the business. / Resource Order Management includes managing the capture of resource orders, scheduling works to support the resource order, managing the fulfillment of resource orders, picking/packing, shipping, tracking and closing orders.

1.5.5.7	L3	Manage Resource Work Order	Manage Resource Order Work business activity directs and controls all work that are required to fulfill an aproved resource order by ensuring the work related to the order is planned, executed and closed in a timely and efficient manner. / Manage Resource Order Work business activity includes activities to "Initiate Resource Work Order", "Create Resource Work Order", "Review Resource Work Order", "Plan Resource Work Order", "Close Resource Work Order", "Analyze Resource Work Order" and "Report Resource Work Order".
1.5.19	L2	Resource Specification Management	[Data-quality issue: this row's Description in the source document is a verbatim copy of activity 1.5.5.7's description, not a description of "Resource Specification Management."] Manage Resource Order Work business activity directs and controls all work that are required to fulfill an aproved resource order by ensuring the work related to the order is planned, executed and closed in a timely and efficient manner. / Manage Resource Order Work business activity includes activities to "Initiate Resource Work Order", "Create Resource Work Order", "Review Resource Work Order", "Plan Resource Work Order", "Close Resource Work Order", "Analyze Resource Work Order" and "Report Resource Work Order".

1.5.19.1	L3	Develop Resource Specifications	Develop Resource Specification business activity define and elaborate on the conditions for meeting resource requirement for consistency, quality, functionality and fit for purpose. / This activity leverages captured resource requirements and resource preferences and builds in tolerances according to stakeholders for suppliers to adhere to.
1.5.19.2	L3	Master Resource Specifications	Master Resource Specifications business activity learn and understand the collection of resource specifications in order to manage improvement or evolution of standards that will used for design and construction of resource. / This business activity is used by organizations to establish a collection of standardized, pre-written specifications or guide specifications that can be used by resource designers, resource architects, and other resource construction entities.
1.5.19.3	L3	Analyze Resource Specifications	Analyze Resource Specifications business activity research, assess and evaluate performance criteria and factors to develop, master, control/update resource specifications for resource requirement adherence. / This business activity can include multiple analyzes of resources specifications, including standard analysis (based on empirical research) or custom analysis (based on other standards). Analysis of Resource Specification include testing Resource specifications and the ability to meet business requirements, as well as conform to operational mandates.

1.5.19.4	L3	Update and Version Resource Specifications	Update and Version Resource Specifications business activity update and track changes to existing resource specifications according to versioning policies to support requirement management. / This business activity manages all changes to resource specifications over their lifecycle. Resource specification versioning is an effective means to communicate changes to resource specifications, so that resource planners and consumers know what to expect. It includes communicating when changes are made to resource specification properties.
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2.3. SID ABEs

SID ABEs this ODA Component is responsible for:

SID ABE L1	SID ABE L1 Definition	SID ABE L2 (or set of BEs)	SID ABE L2 Definition
Workforce	The Workforce ABE models the human and other field resources with their roles, skills, calendar, and other characteristics. It also models the work description and reservation and assignment of a technician to a task. It contains Work, Work Specification, WorkforceResource, and WorkforceSchedule.	Work Specification	The WorkSpecification specifies a kind of work that can be performed manually by a technician. Two key entities that comprise the WorkSpecification ABE are WorkSpecification and WorkSpecFulfillmentRate.
Project	The Project ABE represents the tools used by project managers to ensure that enterprise objectives of quality, cost and time are achieved by planning and scheduling work. It uses common industry definitions of Project, Work Breakdown Structure and Activity to provide support to project managers.	Work Order	<i>(no description available)</i>

2.4. Functional Framework Functions

Function ID	Function Name	Aggregate Function Level 1	Aggregate Function Level 2	Function Description
542	Dynamic Work Order Assigning	Work Order Management	Work Order Assignment and Plan	Dynamic Work Order Assigning takes into consideration the following criteria: // Work order priority / Associated SLAs / Work order due date / Required skills to perform work / Dependencies between work items/orders / Job location relative to other assignments made to the dispatchable staff / Workload of staff / Staff availability hours.
543	Periodic Bulk Work Order Assigning	Work Order Management	Work Order Assignment and Plan	Periodic Bulk Work Order Assigning takes into consideration the following criteria: // Work order priority / Associated SLAs / Work order due date / Required skills to perform work / Dependencies between work items/orders / Job location relative to other assignments made to the dispatchable staff / Workload of staff / Staff availability hours.

545	Work Order Coordination	Work Order Management	Work Order Assignment and Plan	Work Order Coordination function for coordinating work between dependent or related work orders for Resource dispatchable staff.
547	Work Order Information Transferring	Work Order Management	Work Order Tracking and Business Value Development	Work Order Information Transferring provides information transfer between the operator's workforce management system(s) and those of external third parties (when the infrastructure is owned and operated by third parties).
549	Work Order Re-assigning	Work Order Management	Work Order Assignment and Plan	Initiates re-assignments as needed based on changing priorities, technician progress on other job assignments, etc.
540	Work Order Manual Activities Determining	Work Order Management	Work Order Establishment	Work Order Manual Activities Determining function provides the necessary functionality to analyze the work order and determine what manual activities are necessary.
544	Work Order Closing	Work Order Management	Work Order Tracking and Business Value Development	Work Order Closing function closes completed work orders

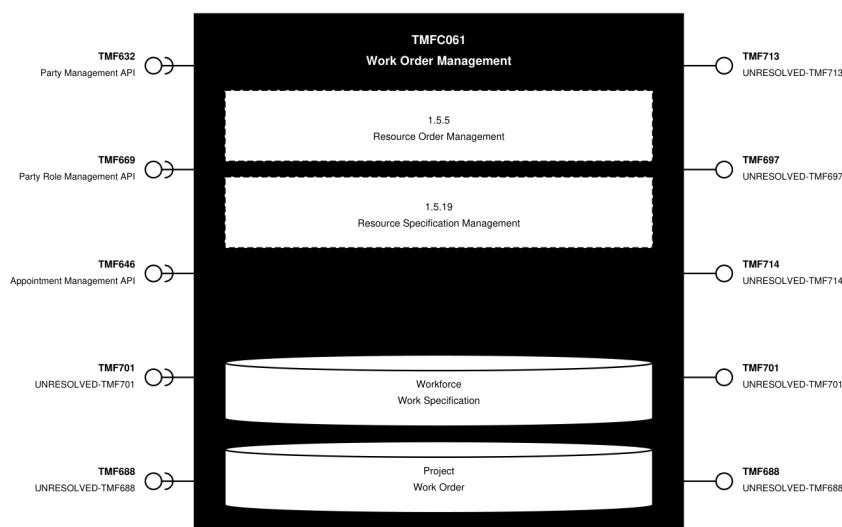
739	Workflow Delay Jeopardy Analyzing	Work Order Management	Work Order Tracking and Business Value Development	Workflow Delay Jeopardy Analyzing function should be found in any process or workflow management system and warns when a delay in the due date of a task or process step will prevent subsequent tasks from starting, or affect the overall end date of the process, also including escalation function for issues as required.
551	Work Order Status Monitoring	Work Order Management	Work Order Tracking and Business Value Development	Monitors the status and progress of the various manual activities, including the various states of the work order
553	Work Specification Classification	Work Specification Management	Work Specification Development	Work Specification Classification provides classification of different types of manual activities provide the necessary functionality to appropriately classify the working activity types and related information.
541	Work Order Assignment Dispatching	Work Order Management	Work Order Assignment and Plan	Dispatching of work order assignments to dispatchable staff.
548	Work Orders Initiation	Work Order Management	Work Order Establishment	Work Orders Initiation initiates workforce orders for Resource preventive and corrective maintenance and change.

552	Work Order Updating	Work Order Management	Work Order Tracking and Business Value Development	Work Order Updating updates the transition of the order through its various states, including in-flight changes and cancellations
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3. TM Forum Open APIs & Events

The following part covers the APIs and Events; this part is split in 3: - List of Exposed APIs - This is the list of APIs available from this component. - List of Dependent APIs - In order to satisfy the provided API, the component could require the usage of this set of required APIs. - List of Events (generated & consumed) - The events which the component may generate are listed in this section along with a list of the events which it may consume. Since there is a possibility of multiple sources and receivers for each defined event.

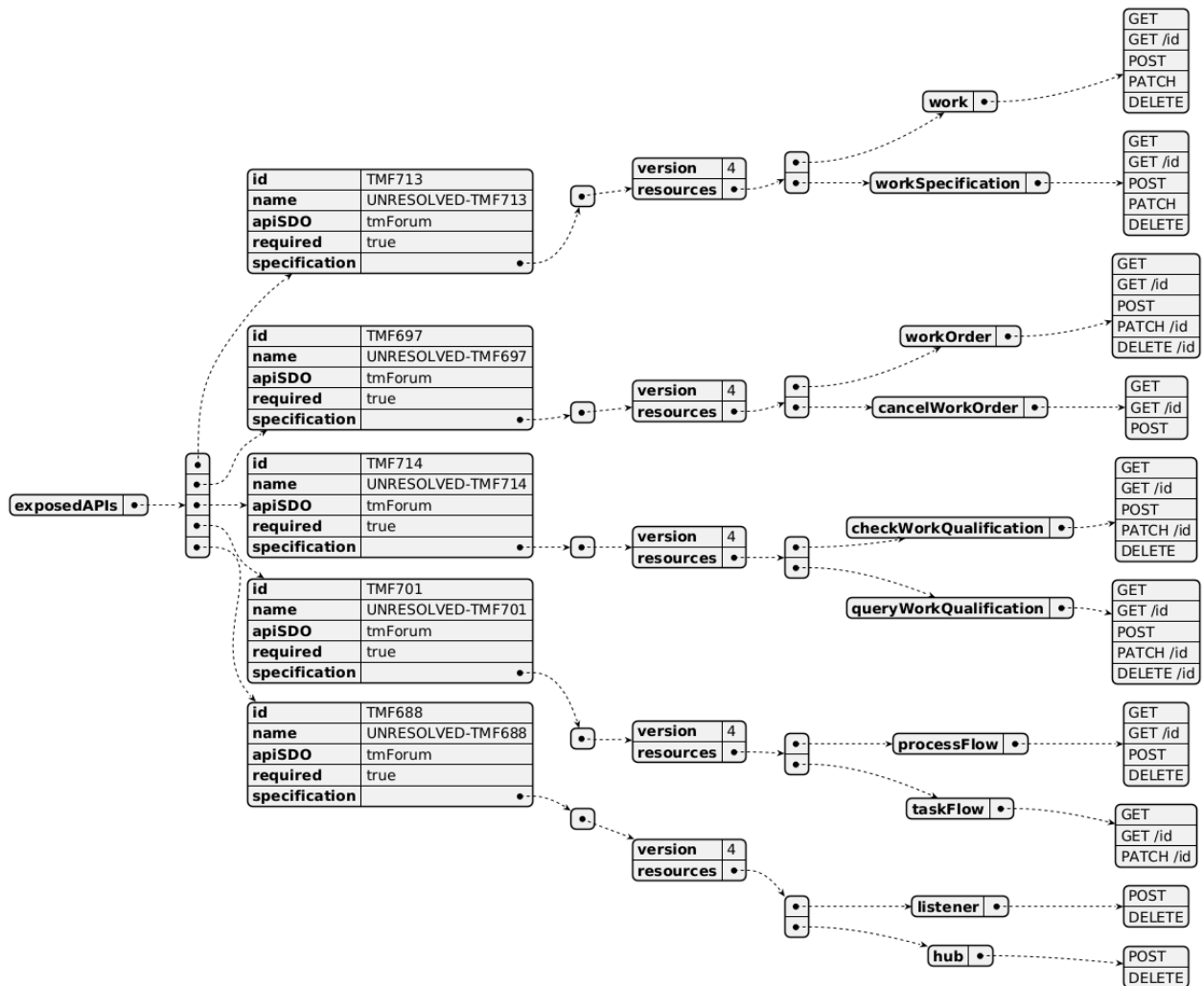
3.1. API Context Diagram



(SVG source: [TMFC061_API_Context.svg](#) — hand-drawn rather than PlantUML for this one diagram, since PlantUML's automatic layout couldn't give each API its own straight, individually-anchored connector at a chosen height. Generated by the component-specification-documentation skill's `scripts/render_api_context_svg.py`.)

3.2. Exposed APIs

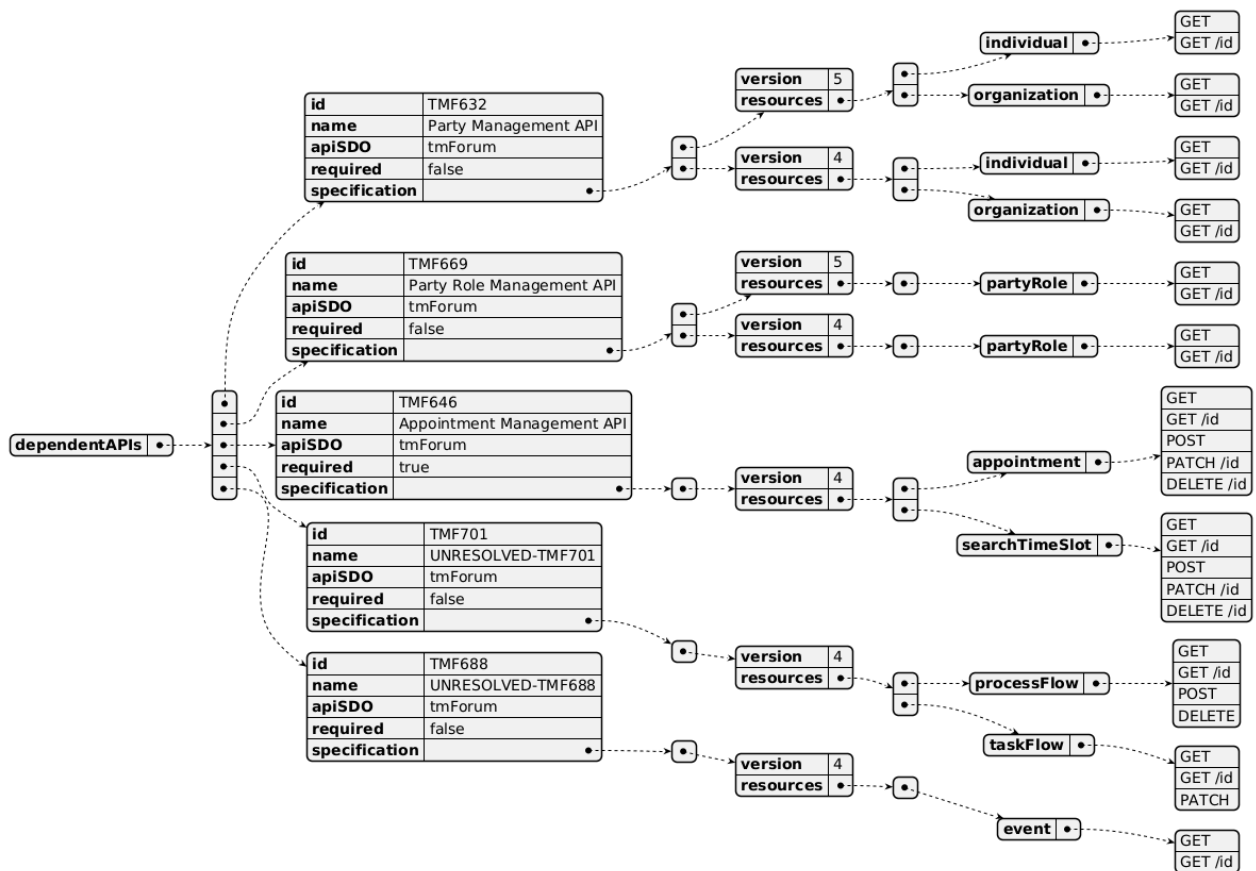
API ID	API Name	Mandatory / Optional	API Version	Resource	Operations
TMF713	UNRESOLVED-TMF713	Mandatory	4	work	GET, GET /id, POST, PATCH, DELETE
TMF713	UNRESOLVED-TMF713	Mandatory	4	workSpecification	GET, GET /id, POST, PATCH, DELETE
TMF697	UNRESOLVED-TMF697	Mandatory	4	workOrder	GET, GET /id, POST, PATCH /id, DELETE /id
TMF697	UNRESOLVED-TMF697	Mandatory	4	cancelWorkOrder	GET, GET /id, POST
TMF714	UNRESOLVED-TMF714	Mandatory	4	checkWorkQualification	GET, GET /id, POST, PATCH /id, DELETE
TMF714	UNRESOLVED-TMF714	Mandatory	4	queryWorkQualification	GET, GET /id, POST, PATCH /id, DELETE /id
TMF701	UNRESOLVED-TMF701	Mandatory	4	processFlow	GET, GET /id, POST, DELETE
TMF701	UNRESOLVED-TMF701	Mandatory	4	taskFlow	GET, GET /id, PATCH /id
TMF688	UNRESOLVED-TMF688	Mandatory	4	listener	POST, DELETE
TMF688	UNRESOLVED-TMF688	Mandatory	4	hub	POST, DELETE



3.3. Dependent APIs

API ID	API Name	API Version	Mandatory / Optional	Resource	Operations
TMF632	Party Management API	5	Optional	individual	GET, GET /id
TMF632	Party Management API	5	Optional	organization	GET, GET /id
TMF632	Party Management API	4	Optional	individual	GET, GET /id
TMF632	Party Management API	4	Optional	organization	GET, GET /id

TMF669	Party Role Management API	5	Optional	partyRole	GET, GET /id
TMF669	Party Role Management API	4	Optional	partyRole	GET, GET /id
TMF646	Appointment Management API	4	Mandatory	appointment	GET, GET /id, POST, PATCH /id, DELETE /id
TMF646	Appointment Management API	4	Mandatory	searchTimeSlot	GET, GET /id, POST, PATCH /id, DELETE /id
TMF701	UNRESOLVED-TMF701	4	Optional	processFlow	GET, GET /id, POST, DELETE
TMF701	UNRESOLVED-TMF701	4	Optional	taskFlow	GET, GET /id, PATCH
TMF688	UNRESOLVED-TMF688	4	Optional	event	GET, GET /id



3.4. Events

The diagram illustrates the Events which the component may publish and the Events that the component may subscribe to and then may receive. Both lists are derived from the APIs listed in the preceding sections.

Published Events

API ID	API Name	Event Resources
TMF713	UNRESOLVED-TMF713	workStateChange workCreateEvent workAttributeValueChangeEvent workDeleteEvent workSpecificationsearchTimeSlotCreateEvent workSpecificationAttributeValueChangeEvent workSpecificationStateChangeEvent workSpecificationDeleteEvent
TMF697	UNRESOLVED-TMF697	workOrderStateChange workOrderCreateEvent workOrderAttributeValueChangeEvent workOrderDeleteEvent workOrderInformationRequiredEvent cancelWorkOrderCreateEvent cancelWorkOrderStateChangeEvent cancelWorkOrderDeleteEvent
TMF714	UNRESOLVED-TMF714	checkWorkQualificationStateChange checkWorkQualificationCreateEvent checkWorkQualificationAttributeValueChangeEvent checkWorkQualificationDeleteEvent checkWorkQualificationInformationRequiredEvent queryWorkQualificationStateChangeEvent queryWorkQualificationDeleteEvent
TMF701	UNRESOLVED-TMF701	processFlowCreateEvent processFlowStateChangeEvent processFlowStateDeleteEvent processFlowStateAttributeValueChangeEvent taskFlowCreateEvent taskFlowStateChangeEvent taskFlowDeleteEvent taskFlowAttributeValueChangeEvent taskFlowInformationRequiredEvent



Subscribed Events

API ID	API Name	Event Resources
TMF632	Party Management API	IndividualCreateEvent IndividualAttributeValueChangeEvent IndividualStateChangeEvent IndividualDeleteEvent OrganizationCreateEvent OrganizationAttributeValueChangeEvent OrganizationStateChangeEvent OrganizationDeleteEvent
TMF632	Party Management API	partyRoleCreateEvent partyRoleAttributeValueChangeEvent partyRoleStateChangeEvent partyRoleDeleteEvent

TMF701	UNRESOLVED-TMF701	processFlowCreateEvent processFlowStateChangeEvent processFlowStateDeleteEvent processFlowStateAttributeValueChangeEvent taskFlowCreateEvent taskFlowStateChangeEvent taskFlowDeleteEvent taskFlowAttributeValueChangeEvent taskFlowInformationRequiredEvent
TMF646	Appointment Management API	appointmentStateChange appointmentCreateEvent appointmentAttributeValueChangeEvent appointmentDeleteEvent searchTimeSlotCreateEvent searchTimeSlotAttributeValueChangeEvent searchTimeSlotStateChangeEvent searchTimeSlotDeleteEvent



4. Machine Readable Component Specification

Refer to the ODA Component Directory on the TM Forum website for the machine-readable component specification files for this component.

5. References

5.1. TMF Standards related versions

Standard	Version(s)
eTOM	v23.5
SID	v25.0
Functional Framework	v23.5

5.2. Jira References

(none currently listed)

5.3. Further resources

1. IG1228: please refer to IG1228 for defined use cases with ODA components interactions.

6. Administrative Appendix

6.1. Document History

6.1.1. Version History

Version Number	Date	Modified by	Description of changes
0.1.0.0	12-Jun-2024	Milind Bhagwat	First draft for team review
1.0.0	02-Jul-2024	Amaia White	Initial release

6.1.2. Release History

Release Status	Date Modified	Modified by	Description of changes
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Pre-production	02-Jul-2024	Amaia White	Initial release.
Production	30-Aug-2024	Adrienne Walcott	Updated to reflect TM Forum Approved status

6.2. Acknowledgements

This document was prepared by the members of the TM Forum Component and Canvas project team.

Team Member	Company	Role
Milind Bhagwat	BT	Editor